POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus I

Semester: Fall

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define continuity and differentiability of a function. Show that Differentiability of a function f(x) at x=a implies continuity but Converse may not be always true. 7

OR

If $\log y = \tan^{-1} x$, show that
i) $(1+x^2)y_{n+2}+(2n+1)x\,y_{n+1}+n(n+1)y_n=0$
ii) $(1+x^2)y_{n+2}+(2nx+2x-1)y_{n+1}+(n^2+n)y_n=0$

b) State and prove Lagrange's Mean value theorem. Interpret it geometrically. 8

2. a) Find the asymptotes of the curve: 8
$x^3+3x^2y-xy^2-3y^3+x^2-2xy-3y^2+4x+5=0$

b) Find the perimeter of the asteroid: $x^{2/3}+y^{2/3}=a^{2/3}$ 7

3. Integrate (**Any Three**) of the following: 3×5

a) $\int \frac{x^2}{(x-2)(x-3)}dx$

b) $\int \frac{1}{4-5\sin x}dx$

c) $\int_0^a \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a-x}}$

d) $\int_0^{\frac{\pi}{2}} \sin^3 x\cos^4 x dx$

4. a) Find the volume of the solid in region bounded by the curve $y=x^2+1$ and the line $y=x+3$ revolved bout the x-axis. 8

b) State and prove Euler's theorem on homogeneous function of two independent variables of degree n. 7

If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+\frac{1}{2}\cot u = 0$.

5. a) Find the extreme values of the function $f(x,y,z)= x^2+y^2+z^2$ subject to the constraints $x+y+z=3a$. 7

b) Show that the substitution $y=y_1+u$ where y_1 is a solution of Riccati's equation, reduces the Riccati's equation to a Bernoulli's equation. 8

6. a) Find the general solution of the differential equation 7
$y''-y'-2y= 3e^{2x}$, $y(0)=0$, $y'(0)= -2$

OR

Solve Second order differential equation of the series RLC circuit
$L\frac{d^2Vc}{dt^2}+R\frac{dVc}{dt}+\frac{1}{C}Vc=\frac{Vin}{dt}$, where
R=10Ω ,L=1 H, C=16×10⁻⁴F V_in=0, Vc(0)=6V, V_c'(0)=6A

b) Find the general solution of the differential equation by using method of variation of Parameters: $y''+9y= \operatorname{cosec} 3x$. 8

7. Attempt all the questions: 4×2.5

a) Find y_n if $y=x^n$,where n is positive integer

b) Find the radius of curvature of the curve $y^2=4x$ at (0,0).

c) Show that the function $f(x,y) = x^3+y^3-3xy$has a saddle point at (0,0).

d) Solve: $\frac{dy}{dx}+\frac{1-\cos 2y}{1-\cos 2x}=0$

CamScanner

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus I

Semester: Fall

Year : 2023
Full Marks:100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) State Leibnitz theorem. If $y=\left(x+\sqrt{1+x^2}\right)^m$, show that 8
i. $(1+x^2)y_2+xy_1-m^2y=0$
ii. $(1+x^2)y_{n+2}+(2n+1)xy_{n+1}+\left(n^2-m^2\right)y_n=0$

b) State Rolle's Theorem and write its geometrical meaning. Verify it for $f(x)=(x-a)^m(x-b)^n$, $x\in[a,b]$ where m and n are positive integers. 7

OR

b) Assuming the validity of the expansion, expand $f(x)=\tan x$ using Maclaurin's series and hence the expansion of $\sec^2 x$.

2. a) Find the asymptotes of the curve $(x-y)^2x^2-a^2(x^2+y^2)=0$. 8
b) Find the radius of curvature at any point of the parametric curve $x=a(t+\sin t), y=a(1-\cos t)$. 7

3. Integrate **Any Three** of the following: 3×5

a) $\int \frac{1}{2-3\sin 3x}dx$

b) Prove that: $\int_0^{\pi/2}\frac{\sqrt{\cot x}}{1+\sqrt{\cot x}}dx=\frac{\pi}{4}$

c) Evaluate: $\int_1^{\infty}\frac{x}{(1+x^2)^2}dx$

d) By using gamma function evaluate $\int_0^{\frac{\pi}{6}}\sin^4 3x\cos^2 6x dx$

4. a) Find the volume of the solid in the region in the first quadrant bounded by the curve $y=x^2$, and the lines $y=0, x=1$ about the line $x=-1$. 8

b) State and prove Euler's theorem of homogenous function of two variables x and y of degree n. 7

If $u=\sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ then show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}\tan u$

5. a) Use Lagrange's multipliers Method, Find the minimum value of $f=x^2+y^2+z^2$ such that $x+y+z=3a^2$ 8

b) Define Bernoulli's equation. Solve: $\frac{dy}{dx}-y\tan x=-y^2\sec x$. 7

OR

b) A tank initially contains 40 kg of salt dissolved into 200 liter of water. A solution of 2 kg of salt per liter is allowed to enter the tank at the rate of 5 liters per minute and uniform solution is drained from the tank at the same rate. Find the amount of salt at any time t. Also determine the salt in tank in 15 minutes.

6. a) Define Euler -Cauchy equation. Solve the initial value problem: 8
$4x^2y''+24xy'+25y=0, y(1)=2, y'(1)=-6$

b) Solve by method of variation of parameter $y''-2y'+y=\frac{e^x}{x^3}$ 7

7. Attempt all Questions. 4×2.5

a) If function $f(x)=\begin{cases}2-x & for\ 1<x<2\\ x-\frac{x^2}{2} & for\ x\geq 2\end{cases}$

Examine continuity of a function at x=2.

b) Check line of symmetry of the curve $y^2(a-x)=x^2(a+x)$.

c) Find the arc length of the curve $x^{2/3}+y^{2/3}=a^{2/3}$

d) Solve: $(1+x)y\,dx+(1+y)x\,dy=0$.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus I

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) If $y=\sin^{-1}x$ then show that: 7
i. $(1-x^2)y_2-xy_1=0$.
ii. $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-n^2y_n=0$

b) State Lagrange's mean value theorem. Also give its geometrical interpretation. Verify this theorem for the function $f(x)=x(x-1)(x-2)$ on $0\leq x\leq\frac{1}{2}$. 8

2. a) Find the asymptotes of the curve: 7
$x^3+x^2y-xy^2-y^3+x^2-y^2-2=0$.

b) i. Find the radius of curvature at any point of the curve 4+4
$x=a(1+\sin t)$
$y=a(a-\cos t)$
ii. Trace the curve of $y^2=(x-2)^3$

3. Attempt (**Any three**) of the following: 3×5

a) Evaluate: $\int\frac{1}{4+5\sin x}dx$

b) Prove that $\int_a^b\frac{\sqrt{x}\,dx}{\sqrt{x}+\sqrt{a-x}}=\frac{a}{2}$

c) Evaluate: $\int_0^{\frac{\pi}{2}}\cos^4 3\theta\sin^2 6\theta\,d\theta$ (apply beta gamma function)

d) Obtain the reduction formula for $\int\sec^n x\,dx$ and hence evaluate $\int\sec^5 x\,dx$.

4. a) State and prove Euler's theorem on homogeneous functions of two variables of degree n. If $\sin u=\frac{x^2+y^2}{x+y}$ then show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\tan u$. 7

b) Find the volume of the solid generated by revolving the region between the parabola $y=x^2$ and the line y=1 about the line y= -1. 8

OR

Find the surface area and volume of the solid generated by revolving the asteroid $x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}$ about x-axis.

5. a) Find the maximum value of $x^2+y^2+z^2$ when $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$ using Lagrange's multipliers method. 7

b) Define Bernoulli's equation. Solve: $\frac{dy}{dx}+x\sin 2y=x^3\cos^2 y$. 8

OR

The growth rate of a culture of bacteria is proportional to the number of bacteria present. After one day it is 1.5 times of original number. Find after how many days it will be
i. double
ii. triple.

6. a) Find the general solution by using the method of variation of parameters: $y''-4y'+4y=\frac{e^{2x}}{x}$ 7

b) i. Solve the initial value problem: 5+3
$(x^2D^2-3xD+4)y=0\ ;y(1)=0, y'(1)=3$.
ii. Find the particular solution of
$y''-y'-2y=2e^{3x}$

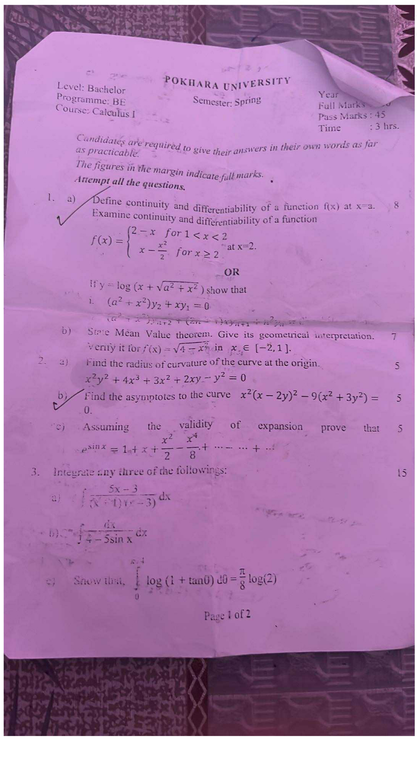
7. Attempt all the questions 4×2.5

a) A function $f(x)$ is defined as follows:
$f(x)=\begin{cases}4-x^2, & x<2\\ x-2, & x\geq 2\end{cases}$
Show that $f(x)$ is not differentiable at $x=2$.

b) Find the area enclosed by x-axis and the curve $y=3x-5x^2$.

c) State Riccati's equation. Also write the condition when it becomes first order linear differential equation.

d) What is the equation of Newton's Law of Cooling?



POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus I

Semester: Spring

Year
Full Marks
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define continuity and differentiability of a function f(x) at x=a. 8
Examine continuity and differentiability of a function
$f(x)=\begin{cases}2-x & for\ 1<x<2\\ x-\frac{x^2}{2} & for\ x\geq 2\end{cases}$ at x=2.

OR

If $y=\log\left(x+\sqrt{a^2+x^2}\right)$ show that
i. $(a^2+x^2)y_2+xy_1=0$

b) State Mean Value theorem. Give its geometrical interpretation. 7
Verify it for $f(x)=\sqrt{4-x^2}$ in $x\in[-2,1]$.

2. a) Find the radius of curvature of the curve at the origin. 5
$x^2y^2+4x^3+3x^2+2xy-y^2=0$

b) Find the asymptotes to the curve $x^2(x-2y)^2-9(x^2+3y^2)=0$. 5

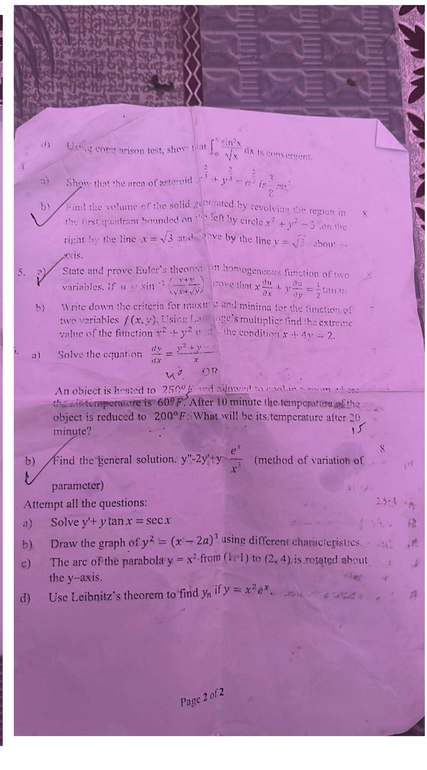
c) Assuming the validity of expansion prove that 5
$e^{\sin x}=1+x+\frac{x^2}{2}-\frac{x^4}{8}+\cdots\cdots+\cdots$

3. Integrate any three of the followings: 15

a) $\int\frac{5x-3}{(x+1)(x-3)}dx$

b) $\int\frac{\sin x}{4-5\sin x}dx$

c) Show that, $\int_0^{\frac{\pi}{4}}\log(1+\tan\theta)\,d\theta=\frac{\pi}{8}\log(2)$



d) Using comparison test, show that $\int_0^{\frac{\pi}{2}}\frac{\sin x}{\sqrt{x}}dx$ is convergent.

a) Show that the area of asteroid $x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}$ is $\frac{3}{8}\pi a^2$.

b) Find the volume of the solid generated by revolution the region in the first quadrant bounded on the left by circle $x^2+y^2=3$ on the right by the line $x=\sqrt{3}$ and above by the line $y=\sqrt{3}$ about y-axis. 8

5. a) State and prove Euler's theorem on homogeneous function of two variables. If $u=\sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}\tan u$.

b) Write down the criteria for maxima and minima for the function of two variables $f(x,y)$. Using Lagrange's multiplier find the extreme value of the function $x^2+y^2+z^2$ with the condition $x+4y=2$. 7

6. a) Solve the equation $\frac{dy}{dx}=\frac{y^2+y}{x^2+x}$ OR

An object is heated to 250°F and allowed to cool in a room where the air temperature is 60°F. After 10 minute the temperature of the object is reduced to 200°F. What will be its temperature after 20 minute? 8

b) Find the general solution. $y''-2y'+y=\frac{e^x}{x^3}$ (method of variation of parameter)

Attempt all the questions: 2.5×4

a) Solve $y'+y\tan x=\sec x$

b) Draw the graph of $y^2=(x-2a)^3$ using different characteristics.

c) The arc of the parabola $y=x^2$ from (1,1) to (2, 4) is rotated about the y–axis.

d) Use Leibnitz's theorem to find y_n if $y=x^2e^x$.